

AKARSHA AGARWAL

Phone: +91 6387671250 | Email: akarshaagarwal25@gmail.com | LinkedIn: [Akarsha Agarwal](#) | GitHub: [Ak-arsha](#)

EDUCATION

Year	Degree / Certificate	Institute	CGPA / %
2023 – 2027	B.Tech, Computer Science & Engineering	Jaypee Institute of Information Technology, Noida	7.17 / 10.0
2022 – 2023	Class XII, CMS Rajendra Nagar 1	Lucknow, Uttar Pradesh	94.75%
2021 – 2022	Class X, CMS Rajendra Nagar 1	Lucknow, Uttar Pradesh	93.8%

EXPERIENCE

Python Developer | [Eglogics Softech Pvt. Ltd.](#)

Mar 2026 – Apr 2026

- Built an automated face-recognition attendance system in Python and OpenCV that identifies individuals via facial features and records attendance digitally, replacing manual tracking.
- Integrated facial recognition with a Flask backend and SQL database, enabling faster, contactless attendance capture and reducing administrative workload and recording errors.

Research Intern | [IIT \(BHU\) Varanasi](#)

May 2026 – Jul 2026

- Developed PaddyCare AI, a deep learning system to classify 10 major paddy crop diseases plus healthy leaves from images, using YOLO and EfficientNet with transfer learning, data augmentation, and fine-tuning.
- Built and deployed an inference pipeline with FastAPI and Streamlit to support early disease detection for farmers, aiding faster intervention and reduced crop loss.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, JavaScript, PHP

Data Engineering & Cloud: PySpark, SQL (PostgreSQL, SQLite), dbt, SQLAlchemy, Pandera, Docker, AWS (S3, EC2, ECR), Dagster, REST APIs, Git/GitHub

ML / DL & CV: PyTorch, TensorFlow, OpenCV, Scikit-learn, XGBoost, LightGBM, MediaPipe, YOLO, EfficientNet, spaCy, Transformers

Web & Frameworks: FastAPI, Streamlit, Plotly, Flask, Django, Django REST Framework, React.js, Node.js, WebRTC

Databases & Tools: PostgreSQL, Supabase, SQLite, Jupyter Notebook, VS Code, Google Colab

Competitive Programming: 1388 rating on Codeforces ([Akarsha__Agarwal](#)) · 300+ problems on LeetCode ([Akarsha11](#)) · 360+ problems on TUF+ ([Akarsha](#))

PROJECTS

DataPulse – Enterprise Multi-Industry Data Platform ([Live Demo](#) | [GitHub](#))

Python, SQL, PySpark, FastAPI, PostgreSQL, Pandera, dbt, Streamlit, Plotly, Docker, AWS, Gemini API, Dagster

- Architected an automated multi-domain ETL data engineering pipeline ingesting 25K+ raw records across Retail, SaaS, Healthcare, and Hi-Tech verticals with Pandera DLQ validation and a 100% row-count reconciliation assertion (*Raw = Clean + Quarantine + Duplicates*).
- Modeled a PostgreSQL Star Schema database with CTEs, Window Functions (*NTILE*), Slowly Changing Dimensions (SCD2), and dbt marts (*v_customer_rfm*, *v_saas_metrics*, *v_healthcare_sla*, *v_hitech_telemetry*) to power executive dashboards.
- Developed a multi-tenant Streamlit + Plotly interactive dashboard with JWT auth/RBAC (Admin, Analyst, Viewer), server-enforced PII masking for healthcare data, and a Google Gemini Text-to-SQL assistant backed by a secured FastAPI REST API.

Picture Perfect – AI-Powered Image Editing Studio ([Link](#))

Python, Streamlit, OpenCV, MediaPipe, Gemini API, Supabase

- Built an interactive web studio for real-time image editing with automated smile enhancement and gaze correction using MediaPipe Face Mesh, plus natural-language photo editing powered by the Gemini API.

KrishiMitra – AI Farm Advisory Platform ([Link](#))

Next.js, FastAPI, XGBoost, LightGBM, Supabase

- Built a full-stack advisory platform for farmers with ML-based 5-day crop price forecasting, an explainable-AI panel, and a geo-spatial buyer recommendation engine using the haversine formula.
- Designed a sell/store decision engine comparing live market prices to government MSP floors, with JWT/Google OAuth authentication and a FastAPI + PostgreSQL (Supabase) backend.

PUBLICATIONS

LocalMark: Robust Localised Message Watermarking for Secure Digital Image Copyright Protection — *Under review, Journal of the Franklin Institute (2026)*

- Co-authored a U-Net-based watermarking framework for embedding patient-identifying messages into medical retinal images; contributed to data curation and manuscript drafting. Proposed Enhanced Robust Decoder achieved 96% bit accuracy and 48.27 dB PSNR across 24 real-world image attacks, outperforming four baseline architectures.

ACHIEVEMENTS

- Selected for the presentation round of Smart India Hackathon 2024 for a solution built around the "Root of Trust" problem statement.
- Selected as Team Lead of the AI/ML Domain under GDG (Google Developers Group), Jaypee Institute of Information Technology.